

Poisson Distribution Exam Revision Guide

Prepared for Eli

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This guide covers all the Poisson distribution concepts, time-scaling tricks, Binomial-to-Poisson approximations, and the exact practice problems we worked through.

The Poisson Core Toolkit

Unlike Binomial (which has a fixed number of trials n), Poisson is used for counting rare, independent events over a fixed interval of time or space at an average rate λ (Lambda).

The Formula

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

Where:

- λ : Mean/average rate of events in the given time/space interval.
- k : The target number of events you want to find.
- e : Euler's constant (≈ 2.71828).
- $k!$: Factorial of k ($k \times (k - 1) \times \dots \times 1$).

Golden Rules & Exam Tricks

1. **Time Scaling:** If the time interval changes (e.g., from per week to per 2 weeks, or per minute to 15 seconds), scale λ directly by multiplying it by the new time factor.
 2. **Binomial Approximation (Large n , Small p):** If $n > 50$ and $p < 0.1$, don't use grueling binomial expansions—approximate using Poisson with $\lambda = n \times p$.
 3. **Complement Rule:** For “at least” or “more than” ($X > k$), use $1 - P(X \leq k)$ to save massive amounts of time.
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Problem 1: Motorway Accidents

Question: Accidents occur on a certain stretch of motorway at the rate of three per month. Find the probability that on a given month there will be: (a) no accidents, (b) one accident, (c) two accidents, (d) three accidents, (e) four accidents.

Variables: $\lambda = 3$ per month. Formula: $P(k) = \frac{e^{-3}3^k}{k!}$

- (a) No accidents ($k = 0$): $P(0) = \frac{e^{-3}3^0}{0!} = e^{-3} \approx \mathbf{0.0498}$ (4.98%)
- (b) One accident ($k = 1$): $P(1) = \frac{e^{-3}3^1}{1!} = 3e^{-3} \approx \mathbf{0.1494}$ (14.94%)
- (c) Two accidents ($k = 2$): $P(2) = \frac{e^{-3}3^2}{2!} = \frac{9e^{-3}}{2} \approx \mathbf{0.2240}$ (22.40%)
- (d) Three accidents ($k = 3$): $P(3) = \frac{e^{-3}3^3}{3!} = \frac{27e^{-3}}{6} \approx \mathbf{0.2240}$ (22.40%)
- (e) Four accidents ($k = 4$): $P(4) = \frac{e^{-3}3^4}{4!} = \frac{81e^{-3}}{24} \approx \mathbf{0.1680}$ (16.80%)

Problem 2: Cretan Airlines Late Arrivals (Time Scaling)

Question: Cretan Airlines services which arrive late to Athens Airport on a typical week can be modelled by a Poisson distribution with mean of 4.5.

a) Determine the probability that on a given week there will be:

- **a. four late arrivals ($k = 4$):**

$$P(4) = \frac{e^{-4.5}(4.5)^4}{4!} \approx \mathbf{0.1898}$$
 (18.98%)

- **b. less than four late arrivals ($X < 4$):**

$$P(X < 4) = P(0) + P(1) + P(2) + P(3) \approx \mathbf{0.3423}$$
 (34.23%)

- **c. at least seven late arrivals ($X \geq 7$):**

$$P(X \geq 7) = 1 - P(X \leq 6) \approx \mathbf{0.1690}$$
 (16.90%)

b) Determine the probability that on a given **two week period** there will be between eight and thirteen (inclusive) late arrivals ($8 \leq X \leq 13$).

- **Scale λ :** New $\lambda = 4.5 \times 2 = 9.0$.
- **Calculation:** Sum $P(8) + P(9) + \dots + P(13)$ using $\lambda = 9.0 \implies \mathbf{0.6023}$ (60.23%).

Problem 3: Estate Agent House Sales (Poisson + Binomial Combo)

Question: Houses sold follow a Poisson distribution with mean 2.5 houses per week.

a) In the next **four weeks**, find the probability of:

- **i. exactly 4 houses ($k = 4$):** Scale $\lambda = 2.5 \times 4 = 10$. $P(4) = \frac{e^{-10}10^4}{4!} \dots$
- **ii. more than 6 houses ($X > 6$):** $1 - P(X \leq 6) = 1 - \sum_{k=0}^6 \frac{e^{-10}10^k}{k!} \approx \mathbf{0.8699}$.

b) In the next **12 of those four-week periods**, find the probability that **exactly 7** periods have more than 6 houses sold.

- **Bridge to Binomial:** $n = 12$ periods, probability of success $p = 0.8699$, target $k = 7$.
- **Formula:** $P(Y = 7) = \binom{12}{7}(0.8699)^7(0.1301)^5 \approx \mathbf{0.0111}$ (1.11%).

Problem 4: Elevator Arrivals (Interval Adjustments & Solving for t)

Question: Elevator arrivals average 4 people every 5 minutes ($\lambda_{\min} = 4/5 = 0.8$ per minute).

(i) **Exactly two people in a 1-minute period** ($\lambda = 0.8, k = 2$):

$$P(2) = \frac{e^{-0.8}(0.8)^2}{2!} \approx \mathbf{0.1438} \text{ (14.38\%)}$$

(ii) **Nobody arrives in a 15-second period:** Scale $\lambda = 0.8 \times 0.25 = 0.2$. $P(0) = e^{-0.2} \approx \mathbf{0.8187}$ (81.87%).

(iii) **Probability of at least one person in next t minutes is 0.9. Find t :**

$$P(X \geq 1) = 1 - P(0) = 1 - e^{-0.8t} = 0.9 \implies e^{-0.8t} = 0.1 \implies t = \frac{\ln(0.1)}{-0.8} \approx \mathbf{2.88} \text{ mins}$$

Problem 5: Machine Rods (Binomial to Poisson Approximation)

Question: 1.2% of rods made by a machine are bent ($p = 0.012$). Sample of 400 rods ($n = 400$).

(i) **State distribution of X :** $X \sim B(400, 0.012)$.

(ii) **Approximate distribution with reason:** $X \approx \text{Po}(4.8)$ (since $\lambda = 400 \times 0.012 = 4.8$).
Reason: Because n is large (400) and p is small (0.012).

(iii) **Probability of more than 2 bent rods ($X > 2$):**

$$P(X > 2) = 1 - [P(0) + P(1) + P(2)] \text{ using } \lambda = 4.8 \approx \mathbf{0.8575} \text{ (85.75\%)}$$

Problem 6: Switchboard Phone Calls

Question: Between 2 PM and 4 PM, average calls per minute is $\lambda = 2.35$. Find the probability that during one particular minute there will be at most 2 phone calls ($X \leq 2$). Given $e^{-2.35} = 0.095374$.

Variables: $\lambda = 2.35$, Target: $P(X \leq 2) = P(0) + P(1) + P(2)$.

$$P(0) = e^{-2.35}(1) \approx 0.09537$$

$$P(1) = e^{-2.35}(2.35) \approx 0.22413$$

$$P(2) = \frac{e^{-2.35}(2.35)^2}{2} \approx 0.26335$$

$$P(X \leq 2) = 0.09537 + 0.22413 + 0.26335 = \mathbf{0.58285} \text{ (58.29\%)}$$

Problem 7: Lake Water Bacteria (Unit Conversion & Scaling)

Question: Lake water contains an average of 3 bacteria per litre (3 per 1000 ml).

a) Sample of 250 ml (0.25 L): $\lambda = 3 \times 0.25 = 0.75$.

• i. **Exactly two bacteria ($k = 2$):**

$$P(2) = \frac{e^{-0.75}(0.75)^2}{2!} \approx \mathbf{0.1329} \text{ (13.29\%)}$$

- ii. At least two bacteria ($X \geq 2$):

$$P(X \geq 2) = 1 - [P(0) + P(1)] = 1 - e^{-0.75}(1 + 0.75) \approx \mathbf{0.1734} \text{ (17.34\%)}$$

- b) Larger sample of 2 litres: $\lambda = 3 \times 2 = 6.0$. Find probability of less than ten but no less than six bacteria ($6 \leq X \leq 9$).

- **Calculation:** Sum $P(6) + P(7) + P(8) + P(9)$ using $\lambda = 6.0 \implies \mathbf{0.4704}$ (47.04%).

Problem 8: Defective Electric Bulbs (Binomial-to-Poisson)

Question: 5% of electric bulbs are defective. Use a Poisson distribution to find the probability that in a sample of 100 bulbs: (i) none is defective, (ii) 5 bulbs are defective.

Variables: $n = 100$, $p = 0.05 \implies \lambda = 100 \times 0.05 = 5.0$.

- (i) None is defective ($k = 0$):

$$P(0) = \frac{e^{-5.0}(5.0)^0}{0!} = e^{-5.0} \approx \mathbf{0.0067} \text{ (0.67\%)}$$

- (ii) 5 bulbs are defective ($k = 5$):

$$P(5) = \frac{e^{-5.0}(5.0)^5}{5!} \approx \mathbf{0.1755} \text{ (17.55\%)}$$